

Regression Algorithms Assignment

K.N.Toosi University of Technology Introduction
to Data Mining

Fall 2024

Implementation Assignment

Dataset

x	y (Label)
-4.5	0
-4	0
-3.5	0
-3	0
-2.5	0
-2	1
-1.5	1
-1	1
-0.5	1
20	1

Table 1: Binary Classification Dataset

Task

This assignment aims to compare the flexibility and accuracy of logistic regression and linear regression in binary classification. You'll fit both models to the given dataset in Table 1 using scikit-learn algorithms, visualize their decision boundaries, and analyze where linear regression falls short compared to logistic regression. Both models use a 0.5 threshold, and accuracy is measured by the number of correctly classified points.

Note

Any attempt to use AI tools for generating the code is strictly prohibited. Students will be asked to present and explain their code during a class session.